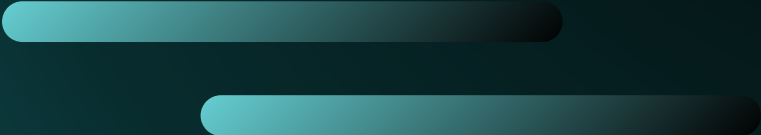


MEDICAL RETRIEVAL-AUGMENTED GENERATION SYSTEM FOR HEALTHCARE KNOWLEDGE ACCESS



INTRODUCTION

- The exponential growth of medical data has made it challenging to access relevant and reliable healthcare information efficiently.
- Traditional search systems lack contextual understanding, while standalone AI models may generate unverified or inaccurate responses.
- To address this, we propose a Medical Retrieval-Augmented Generation (RAG) system that combines semantic search with domain-specific AI to deliver accurate, evidence-based medical insights.
- Traditional search systems like keyword search often return large volumes of irrelevant information, which makes it difficult for doctors to find precise answers quickly.
- Modern Large Language Models (LLMs) can generate natural language responses but sometimes produce hallucinations (incorrect information) because they rely only on their training data.
- To solve this issue, Retrieval-Augmented Generation (RAG) is used. RAG combines information retrieval and generative AI to produce accurate, evidence-based medical answers.

PROBLEM STATEMENT

- The healthcare domain is overwhelmed with massive volumes of medical research, clinical data, and rapidly evolving knowledge, making it increasingly difficult to quickly access relevant, reliable, and up-to-date information.
- Traditional search engines return unstructured and time-consuming results, requiring significant effort to filter and interpret useful insights. At the same time, generative AI models often produce hallucinated or non-verified medical responses, which can be risky in critical healthcare scenarios.
- This creates a significant gap where users lack an intelligent system that can efficiently retrieve trusted medical knowledge and generate accurate, context-aware, and evidence-based insights in real time, ensuring both reliability and usability.

PROPOSED SOLUTION

We propose MedIntel RAG, an advanced Medical Retrieval-Augmented Generation System designed to provide accurate, evidence-based medical knowledge retrieval and explanation.

The system integrates biomedical language models with vector-based semantic search to ensure that every generated response is grounded in real medical literature.

Key Capabilities

- Retrieve relevant medical research papers using semantic search
- Generate context-aware explanations using BioMistral 7B
- Reduce AI hallucinations by grounding responses in verified medical knowledge
- Provide fast and intelligent access to biomedical information

How the System Works

User Query → Semantic Search → Medical Knowledge Retrieval → AI-Powered Explanation

Key Innovation

Unlike traditional AI chatbots, our system first retrieves relevant medical evidence and then generates answers based on that evidence, ensuring higher reliability and transparency.

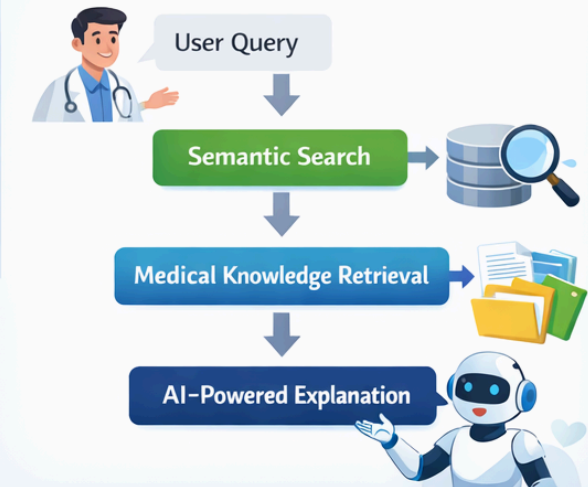
MedIntel RAG

Advanced Medical Retrieval-Augmented Generation System

Key Capabilities

- Semantic Search
- BioMistral 7B Language Model
- Verified Medical Knowledge
- Fast Information Access

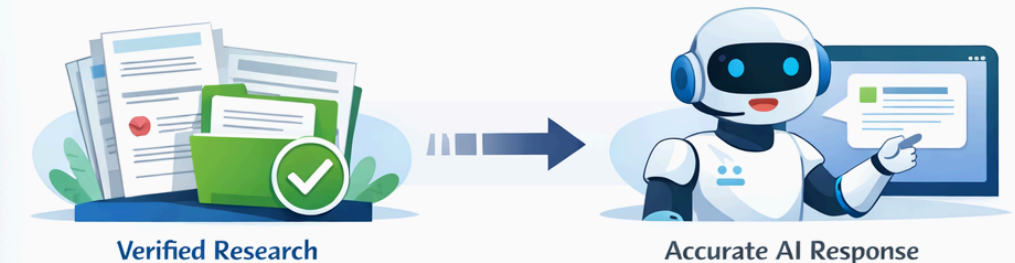
How the System Works



Key Innovation

Evidence First, Then Generate

Retrieve Medical Evidence → Generate Reliable Answers

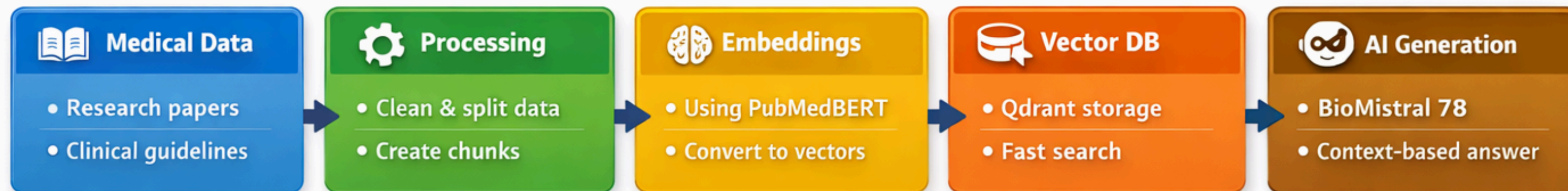


WORKFLOW CHART

System Workflow

Retrieval-Augmented Generation Pipeline for Medical Intelligence

Knowledge Base Creation



Real-Time Query Processing



Evidence First → Then AI Generation

SYSTEM ARCHITECTURE

Step 1: Medical Data Collection

Medical knowledge is collected from trusted sources such as:

- PubMed
- WHO guidelines
- Medical textbooks
- Research journals
- Clinical datasets

Step 2: Data Preprocessing

The collected documents are prepared for processing:

- Cleaning text
- Removing noise
- Splitting documents into smaller chunks

This improves retrieval accuracy.

Step 3: Embedding Generation

Text data is converted into **vector embeddings** using embedding models. Embedding models represent text in **numerical form** so AI systems can compare similarity between queries and documents.

- Sentence Transformers
- OpenAI embeddings
- BioBERT

Step 4: Vector Database

Embeddings are stored in a **vector database**.

- FAISS
- Pinecone
- Chroma
- Weaviate

These databases allow **fast similarity search**.

Step 5: Retrieval Module

When a user asks a question:

1. The query is converted into an embedding.
2. The vector database searches for **similar documents**.
3. The most relevant medical knowledge is retrieved.

Step 6: Large Language Model (LLM)

The retrieved documents are passed to an LLM such as:

- GPT
- LLaMA
- Med-PaLM

The LLM generates an answer based on the retrieved context.

This ensures the answer is **fact-based and reliable**.

TECHNOLOGY STACK

01 Programming Language
Python

02 AI Frameworks

- LangChain
- LlamaIndex

03 Embedding Models

- Sentence Transformers
- BioBERT

04 Vector Databases

- FAISS
- Pinecone
- Chroma

05 Large Language Models

- GPT
- LLaMA
- Med-PaLM

06 Frontend

- React
- Streamlit

Impacts Benefits of Medical RAG Systems

1. Accurate Medical Information

Answers are generated using verified medical sources.

2. Reduced AI Hallucinations

The model relies on retrieved documents rather than only training data.

3. Faster Knowledge Access

Doctors can quickly find medical information during clinical practice.

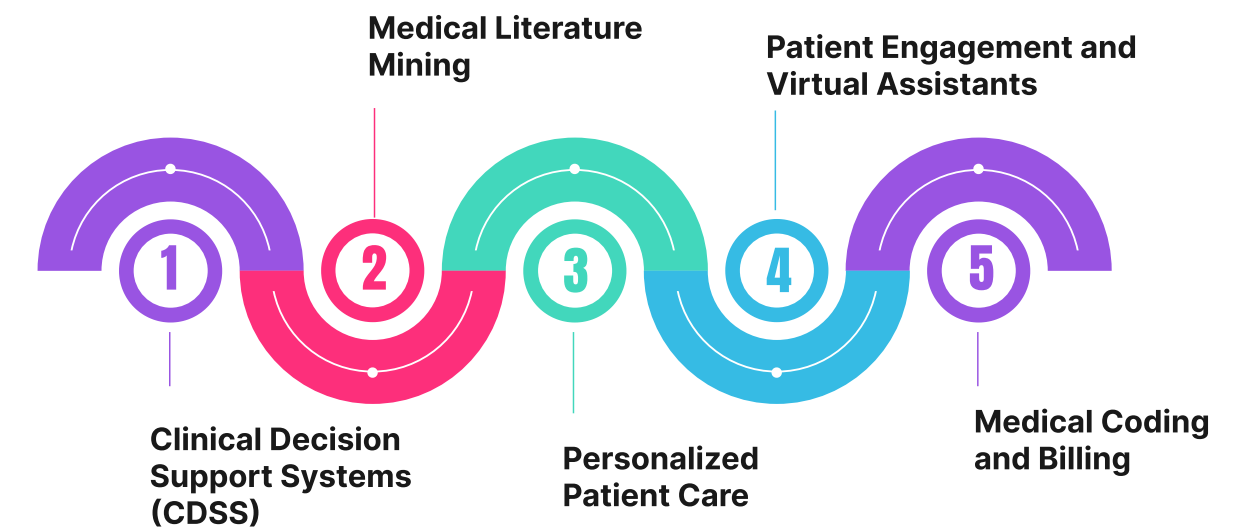
4. Evidence-Based Medicine

Responses are grounded in scientific research.

5. Support for Medical Education

Medical students can use the system for learning and research.

Key Use Cases of RAG in Healthcare



Future Scope

The system can be extended into a clinical decision support tool to assist doctors with real-time, evidence-based recommendations.

- Integration with real-time hospital data and electronic health records (EHRs) can enable personalized medical insights.
- Expansion to include more trusted datasets such as World Health Organization guidelines and global clinical research for broader coverage. Development of a multilingual interface to make medical knowledge accessible to users in regional languages.
- Addition of voice-based interaction to support hands-free usage for healthcare professionals and patients.
- Incorporation of explainable AI features, showing sources and reasoning behind each generated response.
- Scaling the system into a mobile or web application for widespread public and professional use.
- Integration with wearable devices for real-time health monitoring and intelligent suggestio

Thank You